VOLLEY | Presenter Reference

Slide notes, key numbers, equations, and viva preparation

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Guide: Vikas Gulia **Student PRNs:** Adityavardhan Mishra — 23070125054; Pratham Chawla — 23070125029 **Carry with:** main PPTX and panel handbook **Purpose:** private cue book for both presenters; it is not a script to read word-for-word.

> **Opening thesis statement:** "We completed a system-level computational study of a commandable CubeSat deployer. It predicts an orbital benefit but fails our preset 3U mass comparison. The Gen5 research question has an answer; physical qualification and Gen6 belong to the next phase."

Rehearsed opening and close

Opening (about 35 seconds): "Our thesis asks whether a hosted deployer can give each rideshare CubeSat a commanded release condition that improves its orbit enough to justify the whole machine. We completed a Gen5 computational evaluation. The historical model predicts 16.0 m/s, while a finite geometry force screen challenges that performance claim. The mass comparison fails, the matched twelve-shot screen closes no case, and a native CAD review identifies an 11 mm packaging shortfall in our side-fed reference layout. We will show those checks and the engineering decision."

Close (about 25 seconds): "The research question now has documented negative findings and an unresolved force-model discrepancy. A commanded release has conditional orbital value in the model, but this Gen5 configuration is not competitive on the stated 3U mass criterion. That finding closes the Gen5 analytical study. Physical qualification and the Gen6 mechanism trade are future work."

Aim for an 11-minute full route in rehearsal, then shorten to the key slides if the panel gives a tighter slot. The numerical claims and failure criterion should remain in either route.

Before entering the room

- Bring the main PPTX, a PDF export of the same slides, the panel handbook PDF, and this reference PDF on two devices/USB copies if possible.
- Confirm the projector can read the title, units, figure axes and footer caveats. Use the PDF export if animations or fonts shift in PowerPoint.
- Open the model result files or screenshots for the simulation demonstration. Do not imply a live physical demo.
- Decide which student covers each section. A natural handoff is after **Timeline** (slide 10) or after **Methodology** (slide 8); either student should be ready to answer questions across the whole project.

- If the review slot is still unknown on arrival, confirm it before starting and use the rehearsed full or short route accordingly.

Short route if time is limited

Prioritize slides **1, 3, 5–7, 9–14, 17–18, 19, 21–23**, keeping the literature and model tables brief. Lead with the thesis answer and close on the engineering decision. Keep the mass failure and evidence boundary. Slides 25–26 are references for questions. The full route follows all 26 slides in order.

Slide-by-slide speaking notes

1. Title

"We are Adityavardhan Mishra and Pratham Chawla, guided by Vikas Gulia. We completed a system-level computational evaluation of controlled CubeSat deployment. The study predicts useful orbital change but finds that our Gen5 design fails the 3U mass target. We will show the design, methods and result before briefly outlining Gen6 future work." The cover image is the university logo. The STEP-derived views beginning on slide 3 are model geometry, not physical hardware.

2. Outline

Follow the university-requested order exactly: introduction, literature, gap, objectives, methodology, timeline, progress, results, conclusion, future work and references. Explain that results include failures and limitations because these are part of the engineering decision.

3. Introduction

NASA documents the rideshare destination constraint; Planet and Spire show repeatable smallsat platforms. Those are market context, not VOLLEY customers. A carrier, rideshare integrator or fleet operator is only a candidate buyer for several useful release states from one host. The image is the Gen5 STEP reference with its enclosure hidden; its side-fed fit fails. Compare actual spring impulse, host manoeuvres, transport and onboard propulsion on the same mission. Waiting alone between zero-relative-impulse releases from an unchanged host creates no persistent phase separation. Say the finding early: modeled Gen5 orbital benefit is conditional, while its 3U mass comparison fails. There is no confirmed buyer or approved host. The local MARKET_AND_CUSTOMER_FIT.md audits the source reports and spacecraft classes.

4. Literature review

Do not describe springs as "obsolete" or "unable to change orbit." Every impulse changes the orbit. Springs are the benchmark for simplicity and heritage. Spring impulse and differential drag address some phasing needs. Waiting alone between zero-relative-impulse releases from an unchanged host does not. A transfer vehicle addresses larger moves but carries its own burden. Electromagnetic work predates VOLLEY: Feng *et al.* models high-speed on-orbit launch, and Zhao *et al.* studies stacked CubeSat transfer/storage. These works narrow the novelty claim.

5. Research gap

Say: "Our contribution is the coupled, installed-system question: commandable release for an unmodified satellite at moderate acceleration, with magazine, power, brake, host interaction and mass counted." Do not claim the first electromagnetic deployer. "Unmodified" is a design objective, not proof that every payload is compatible; the magnetic and mechanical interface still needs qualification.

6. Objectives

Walk down the outcome column as completed research tasks: the architecture was modelled; shot, circuit and orbital results were calculated; installed mass was counted; key numerical claims were challenged. The mass criterion failed, which is itself a finished result. If asked whether objectives were achieved, distinguish completing a research evaluation from proving hardware performance. The latter was outside this computational thesis's evidence.

7. Research methodology — flow

Describe the completed chain in past tense: we chose the 3U case; sized geometry and magnet sled; calculated force, pulse and release state; propagated the orbit; compared orbital value and installed mass with springs; and challenged models through independent numerical checks. The checks changed the final baseline, including the sled mass and magnetic thrust integral.

8. Research methodology — evidence

Summarize what the completed checks established and their scope. Field agreement at a tested location is not an independent thrust measurement. ngspice checks circuit behavior under assumed components. CalculiX checks a defined structural boundary case. OpenFOAM checks aerodynamic behavior, not vacuum release contact. GMAT and alternate propagation show that atmosphere matters. Use **numerical verification** or **cross-check** for these, and reserve **experimental validation** for physical measurements.

9. Timeline — past

Emphasize documented decisions rather than claiming all early dates were exact. The host concept is documented in 2023. The move toward a linear synchronous motor is approximately mid-2025. CAD and model development spans 2025–26. July–September 2026 focused on cross-checks, corrections and the thesis record. The final review presents that completed Gen5 analysis.

10. Timeline — corrections in the final baseline

This is the strongest proof that the thesis is a finished analysis rather than an early proposal. CAD-derived sled mass displaced an optimistic parametric estimate; a depth-resolved thrust integral lowered the historical model velocity; a real enclosure buildup exposed the mass failure; and an independent propagator check rejected an overbroad solar-activity claim. State that the old shot numbers remain traceable, but the finite-force discrepancy prevents a final speed rating and that failed claims were retained in the record. Do not present these as physical validation.

11. Work progress

Show tangible completed deliverables: manuscript, CAD, system model, scripts, numerical run sheets, baseline, provenance and defect register. The STEP-derived outer-envelope view conceals the documented internal clash; point that out before anyone mistakes it for an approved package. State that the thesis closes an analytical question; hardware tests, a qualified payload and host agreement are separate development milestones. Avoid a percentage-complete claim, since research completion and product readiness are different dimensions.

12. Results — architecture

Explain the intended chain in the STEP-derived view: cassettes would feed a 3U satellite; the reusable magnet sled moves to a release station 1.5 m from the breech along 1.8 m structural longerons; the payload would depart and the sled be arrested. The image shows the fixed reference placement, not a modeled working feed sequence; its enclosure is hidden and the side-fed solids clash. Eight source STEP B-reps were imported into FreeCAD 1.0 and exported as a native 20-instance review assembly and STEP exchange files. The imported solids lack parametric feature history. This is an inspectable geometry study, not an independently mass-verified installed assembly. The mass model still adopts a 9.45 kg sled from historical Gen3 CAD and A4 chassis analysis. No component has been built for VOLLEY.

13. Results — historical modeled shot

At the historical modeled 3U point, the periodic-force model predicts 16.029 m/s and 10.07 g. Gross electrical draw is 2,782 J, of which the payload receives about 514 J as kinetic energy. Only 47 J is recovered after release within the available 39 mm zone; 1,162 J of sled energy goes to the brake. Net electrical-to-payload efficiency is 18.8%. Peak current is 320 A over a 162.3 ms pulse. P117 checks simple identities and reconciles **124.488 J of compact-output remainder** to assumed converter, auxiliary and integration terms. P119's 2-D FEM gives 1.082 kJ in-plane work; P121 independently gives 1.042 kJ ideal full-depth work; P122's illustrative 14 mm gap reduces it to 0.907 kJ. P120's assumed full-winding bank case gives 2.099 kJ gross under the finite force curve. Neither selects hardware or verifies a release speed. Do not say "tested," "measured," or "qualified."

14. Results — finite force discrepancy

Explain that 16.029 m/s assumes periodic force through the whole 1.3 m. The CAD array loses direct stator overlap after 1.066 m. A 3-D analytic volume integral of the finite belts gives 1.042 kJ of ideal work and 12.448 m/s with phase optimized at every point; real circuit limits reduce this model result. This screen shares the earlier magnetic field law, so independent FEM force verification remains open. Do not call 12.448 m/s measured or an accepted replacement baseline.

15. Results — command precision

The closed-loop simulation produces a 0.0274 m/s (3σ) spread around a 15.8 m/s setpoint. It includes assumed sensor and plant tolerances, with control headroom. A panel member may ask whether that precision survives real contacts, thermal changes or payload variability. Answer: those are unmeasured and require an instrumented release test and model correlation.

16. Results — orbit

For the assumed historical 16.029 m/s release state, an independent Cartesian two-body integration reproduces the immediate 28.800775 km semi-major-axis rise within the declared numerical band. This checks orbital mechanics for that state, not atmospheric lifetime. The model's 1.60× lifetime multiplier at mean solar activity remains unverified at this exact point; a prior invariance claim failed an independent propagator check. Spring impulse, differential drag or a host maneuver can create relative state and phase satellites; waiting alone cannot.

17. Results — mass

This is the decisive negative result. Dry mass is 126.6 kg, loaded mass 174.6 kg. The twelve-3U configuration spends 10.55 kg of deployer per satellite, about 1.76 times the roughly 6 kg spring-canister comparison. The predeclared criterion was within 15%, so Gen5 fails. A modelled 50.04 kg enclosure replaced an 8 kg placeholder and exposed the penalty. Do not imply the full assembly is mass-competitive for 3U.

18. Results — CAD packaging check

Show the dimensioned section from the FreeCAD review assembly. Its 526 mm inner enclosure width cannot accommodate the 205 mm track and two 166 mm side cassettes: 537 mm total, an 11 mm shortfall before clearances. Both cassettes intersect the track by 32,915 mm³ in this exact placement. This fails the **side-fed reference layout**; it does not rule out every layout or determine a manufacturable tolerance stack. It is a real negative CAD check, not a rendering defect. P116 records the geometry and reproduction steps.

19. Results — unselected R1 feeder

The widened 570 mm enclosure clears twelve scripted 3U-envelope routes in exact-solid geometry. The slide's STEP-derived Blender view hides that enclosure; it is a candidate arrangement, not a moving simulation or FreeCAD GUI screenshot. Show the native FreeCAD file and STEP set as review artifacts. The real lift/carriage and launch retention have not been designed, and R1 needs new mass, host and structural analyses. It is separate from the evaluated Gen5 configuration.

20. Results — simulation demonstration

Walk through the revised evidence chain on the slide: P118's finite-position force curve challenges the historical constant-force shot, P119 independently checks the 2-D field trend, P121 separately checks the ideal full-depth integral, P122 tests illustrative geometry perturbations, and P120 reruns the bank trajectory conditionally. The common-input mission screen compares the assumed finite-force ideal upper speed with springs and the historical model. The accepted twelve-shot prefixes are limited and no case closes. The panel can inspect P118–P122, the matched-mission result JSON, scripts, FreeCAD documents and reports in this repository. If a script cannot run during review, the slide and handbook reproduce the captured outputs. Call this a **computational demonstration**, never a motor test.

21. Results — matched reference mission

The comparison fixes one payload, host, start orbit and target. With assumed device masses, spring release needs 2.693 kg ideal host propellant for one matched target; finite Gen5 needs 0.827 kg. **The P123 card now adds the installed comparison:** after resizing fuel for only that event and equal reserve, Gen5 device plus fuel is still 52.717 kg heavier. The common-input twelve-shot optimizer finds accepted prefixes 4/12 and 1/12 respectively;

neither closes. The optimizer result and component masses are scenario assumptions, not a customer benefit or a proof of impossibility.

22. Conclusion

"We documented Gen5's computational results and decisive checks. The finite-force screen challenges the historical speed prediction, the modeled configuration fails our 3U mass criterion, and the evaluated side-fed geometry clashes. None of the matched twelve-shot reference cases closes. Gen5 has recorded adverse results and open force closure, not a selected flight configuration." Future work follows separately.

23. Future work — Gen6

Gen6 retains the hosted, sequential-delivery mission and reopens hardware selection. The updated slide shows **P130's matched four-release surrogate**, not measured delivery: both commanded release and spring-plus-host reach the same ideal full-state targets, while assumed installed device plus consumed host fuel favors springs by 10.7177 kg. A +0.1 m/s release error creates about 25–27 km position difference after one day in this model. The first article has two partial FreeCAD/STEP envelope candidates; P131–P132 leave motor, source, contact, brake and full installed mass open. The 1 km/s objective remains distant research; at 1.3 m and 4 kg it would imply about 39,220 g ideal mean acceleration. No achieved command range or product benefit is asserted.

24. Future work — test article

The immediate target is to close G0 customer/host/interface requirements and G1 complete architecture selection. The two partial candidates do not yet have selected circuits, rated source, release contact, launch restraint, carrier capture/brake or an installed bill of materials. A later build package needs controlled geometry, interfaces, mass/energy/thermal budgets, calibration and instrumentation, pass/fail and stop criteria, uncertainty and retained failures. A configuration may be rejected. Physical manufacture, testing and qualification are later milestones.

25–26. References

Offer the panel handbook for the fuller bibliography. Distinguish published/agency sources from our own project manuscript and PLAN-R2. If a panel member asks for a numerical source, give the relevant baseline, figure and run sheet rather than citing "the internet."

Key numbers card

Quantity	Defensible phrase
16.029 m/s	"Historical periodic-force model prediction, challenged by finite geometry"
12.448 m/s	"Finite analytic geometry-only speed under ideal phase and omitted circuit losses"
10.07 g	"Modelled payload acceleration, not payload qualification"
2,782 J	"Gross modelled bank draw per shot"
47 J	"Modelled recovered energy in available zone"
1,162 J	"Modelled sled energy dissipated by brake"
18.8%	"Net modelled electrical-to-payload efficiency"
0.0274 m/s (3σ)	"Closed-loop simulated velocity dispersion"
126.6 / 174.6 kg	"Gen5 modelled dry / loaded mass"
10.55 versus ~6 kg	"Gen5 versus spring mass per 3U customer"
28.8 km	"Predicted semi-major-axis rise for the stated shot"
1.60×	"Predicted lifetime at mean solar activity, conditional"
1–2 m/s to 1 km/s	"Gen6 investigation envelope, not achieved performance"

Equations to recall

- Payload kinetic energy: $E_k = \frac{1}{2}mv^2$. With $m = 4$ kg and $v \approx 16$ m/s, $E_k \approx 512$ J (baseline: 514 J).

- Mean acceleration over a constant-acceleration approximation: $a \approx v^2/(2s)$. It is a sanity check, not a substitute for the modelled time-varying force and the 10.07 g reported result.
- Release momentum/recoil is approximately $p = mv$ for the payload; the complete host impulse must also include sled and actuator dynamics. The thesis baseline reports 64.1 N · s recoil per shot for its defined case.
- Mass per 3U customer: $126.6/12 = 10.55$ kg. Against ~6 kg, the ratio is ~1.76; the target band was ±15%.
- Electrical-to-payload efficiency: payload kinetic energy divided by **net** electrical draw, $514/(2782 - 47) \approx 18.8\%$.

Likely viva questions

What is genuinely new? Not electromagnetic launch itself. The thesis's narrow contribution is a commandable moderate-acceleration release of an unmodified CubeSat with the whole magazine, power, brake, mass and orbital-use case analysed together. Prior art narrows even this, so it is a research proposition, not a proven patent claim.

Why not just release satellites at different times? Different release times can matter if the host maneuvers, spring impulses differ, or differential drag accumulates. Waiting alone between zero-relative-impulse releases from an unchanged host does not create persistent in-track spacing. A conventional spring release is the baseline; a larger directed impulse may change semi-major axis more quickly.

Why not put propulsion on each CubeSat? That is a competent alternative, especially at 3U. The thesis mass comparison even shows Gen5's per-customer deployer mass is unfavorable for 3U. The value proposition may vary by payload class and mission; it is not assumed.

What is the strongest result against your own idea? The full 3U mass rollup: 10.55 kg per customer against about 6 kg for the spring comparator. The preset 15% parity criterion fails.

What has actually been validated? No hardware. Some models have independent numerical checks at particular points and boundary conditions. We should call those verification/cross-checks and state exactly what they do not test.

Can an unmodified CubeSat withstand 10 g and the magnetic field? That cannot be assumed. Structural, shock and magnetic-cleanliness requirements are payload-specific and would need a named payload/interface, testing and provider approval.

What about a failed reload or latch? Shared mechanisms can threaten later releases. The defect and reliability analyses identify that exposure; cycle-life and fault-recovery measurements are needed before claiming reliability.

Is 1 km/s achievable by Gen6? No such result exists. It is an upper research target for screening architectures. Travel length, acceleration, energy, loads, thermal behavior and provider constraints may reject it.

Has a provider agreed to host VOLLEY? No. Published host examples are parametric worked cases. Compatibility requires provider interface data and agreement.

What is the next experiment? First freeze the selected mission and configuration. Then use a discriminating instrumented coupon to measure the mechanism's force/contact/release behavior and compare it with predeclared acceptance and stop criteria. The exact coupon follows the Gen6 trade, not the other way around.

Phrases to avoid in the review

Avoid	Say instead
"We built/tested VOLLEY"	"We designed and analysed Gen5; no hardware test has occurred."
"Validated 16 m/s"	"The historical periodic model predicts 16 m/s; finite-force checks challenge it, and no selected speed rating exists."
"CubeSats can tolerate 10 g"	"Payload-specific qualification has not been done."
"VOLLEY is lighter than springs"	"Gen5 fails the 3U spring-canister mass comparison."
"Gen6 reaches 1 km/s"	"Gen6 must investigate up to 1 km/s; no mechanism is selected."
"ISRO/Skyroot is our partner"	"They were worked host examples, without approval or integration."
"We invented electromagnetic CubeSat deployment"	"The literature contains related electromagnetic approaches."

Source quick map

- Main results and narrative: `../../source/paper.tex` at local head `b30ffdc` (28 September 2026).
- Rated values: `../../appendix/BASELINE.md`.
- What is and is not checked: `../../appendix/PROVENANCE.md`.
- Design-critical open issues: `../../appendix/OPEN_PROBLEMS.md`.
- Historical milestone confidence: `../../appendix/HISTORY.md`.
- Gen6 target and gates: `CURRENT PLAN AND STATUS - PLAN-R2`, 29 September 2026.
- University scoring: attached `B-Tech Docs required.pdf`.